

Assignment 2

Back Ground: The dataset contains 5,000 PMSGMBY beneficiary application records from 20 states and 40+ DISCOMs covering applications filed between January 2024 and May 2025. Each row represents one application.

Key scheme rules embedded in the data:

- CFA = 60% of benchmark cost for up to 2 kW; 40% for the 3rd kW; capped at ₹78,000
- Benchmark cost: ₹50,000/kW (general states); ₹55,000/kW (special category — hilly & NE)
- DBT SLA: CFA should reach beneficiary within 30 days of commissioning
- Bank loan mandate: Interest rate $\leq 7\%$ for collateral-free solar loans
- Grievance SLA: Resolution within 30 days of raising
- ALMM: Approved List of Models and Manufacturers — installations must use listed equipment

Python Analysis

All questions should be answered in a single file. Candidates are required to submit answers to all questions in a single file. Relevant Python scripts/notebooks must also be provided.

Data Cleaning & Exploration

Q#	Question	What to Evaluate
A1	Load the CSV and display basic dataset info: shape, column types, null counts, and % missing per column.	pandas proficiency; ability to assess data quality before analysis
A2	Standardise date columns (Application_Date, TFC_Date, etc.) to datetime. Calculate a new column Total_Pipeline_Days = CFA_Credit_Date – Application_Date. Plot a histogram of this column for commissioned applications.	Date parsing, feature engineering, distribution visualisation
A3	Identify and handle outliers in Market_Cost_Rs. Define and justify your outlier threshold. Show a box plot before and after treatment.	Statistical thinking, outlier detection, judgement on treatment strategy
A4	Check data integrity: are there applications where CFA_Sanctioned_Rs exceeds the scheme maximum (₹78,000)? Are there records where Actual_Monthly_Units > Expected_Monthly_Units × 1.5? Flag and report these anomalies.	Domain-knowledge-driven QC; ability to apply business rules programmatically
A5	Compute summary statistics (mean, median, std, min, max, percentiles 25/75) for: TFC_Processing_Days, CFA_Credit_Days_After_Commission, Monthly_Bill_Savings_Rs, and Satisfaction_Score. Present as a formatted summary table.	Descriptive statistics; pandas describe / agg; formatting skills

Financial Architecture Analysis

Q#	Question	What to Evaluate
B1	Calculate the Market Cost Gap % = $\frac{(\text{Market_Cost_Rs} - \text{Benchmark_Cost_per_kW} \times \text{System_Size_kW})}{(\text{Benchmark_Cost_per_kW} \times \text{System_Size_kW})} \times 100$ for each record. What % of applications have a market cost MORE than 20% above benchmark? Visualise with a histogram.	Core scheme evaluation: is the ₹50,000/kW benchmark still market-accurate?
B2	Among applicants who took loans, compute the distribution of	Financial inclusion check; state-level equity analysis

	Actual_Interest_Rate_pct. What % were charged above the mandated 7%? Break down by State. Plot a bar chart of states sorted by % of above-mandate interest rates.	
B3	Calculate the actual DBT lag (CFA_Credit_Days_After_Commission) for all commissioned beneficiaries. What % received CFA within the 30-day SLA? Produce a DISCOM-level compliance table (% within SLA, avg lag days) and identify the top 5 worst-performing DISCOMs.	Fund flow efficiency; DISCOM performance benchmarking
B4	Compute the average Actual Monthly Bill Savings (Rs.) segmented by: (a) System_Size_kW and (b) Beneficiary_Category. Is the scheme delivering proportionally more benefit to BPL / SC/ST households? Plot a grouped bar chart.	Impact equity assessment; scheme objective alignment
B5	DISCOM Incentive Utilisation Rate = $\text{DISCOM_Incentive_Utilised_Rs} / \text{DISCOM_Incentive_Received_Rs}$. Compute this for each DISCOM. Plot a ranked horizontal bar chart. Which DISCOMs are underutilising incentives? Correlate utilisation rate with the average TFC_Processing_Days for that DISCOM.	Programme accountability; correlation between incentive use and performance

Implementation Pipeline Analysis

Q#	Question	What to Evaluate
C1	Build a pipeline funnel analysis: out of 5,000 applications, how many reached each stage (TFC, Installation, Commission, CFA Credit)? Calculate the drop-off % at each stage. Visualise as a funnel chart or horizontal bar chart with drop-off annotations.	End-to-end pipeline efficiency; critical bottleneck identification
C2	Compute the average days at each pipeline stage (TFC_Processing_Days, Installation_Days_After_TFC, Commission_Days_After_Install, CFA_Credit_Days_After_Commission) by State. Which state has the longest total average pipeline time? Create a stacked bar chart.	State-wise processing efficiency; target for operational improvement
C3	ALMM Compliance rate = % of commissioned installations using ALMM-compliant equipment. Compute state-wise. Which states have <75% compliance? Is there a correlation between ALMM compliance and Vendor_Rating?	Regulatory compliance tracking; vendor quality linkage

C4	Grievance Analysis: Among applications with a grievance, compute (a) Grievance type distribution, (b) Resolution rate per type, (c) % resolved within 30-day SLA per type. Plot two charts: a pie chart of grievance type volume and a bar chart of resolution rates.	Grievance mechanism effectiveness; highest-burden categories
C5	Build a simple linear regression model to predict Satisfaction_Score using: System_Size_kW, Monthly_Bill_Savings_Rs, Vendor_Rating, Grievance_Resolved (1/0), and Total_Pipeline_Days. Report R ² , coefficients, and interpret which factor most influences satisfaction.	Predictive modelling; feature interpretation; beneficiary experience drivers

Equity & Impact Analysis

Q#	Question	What to Evaluate
D1	Compute the share of each Beneficiary_Category in the dataset. Compare BPL + SC/ST + Women-headed combined share with the national population share for these groups (assume ~55% of population). Is the scheme reaching the intended marginalised beneficiaries?	Equity and inclusion assessment; Jan Bhagidari evaluation
D2	Compute state-wise installation counts (records that reached Commission_Date). Calculate the Gini coefficient of the state-wise installation distribution. Visualise with a Lorenz curve. Which 3 states account for >50% of all installations?	Geographic concentration analysis; Module 4.3 of evaluation framework
D3	Performance Efficiency Ratio = Actual_Monthly_Units / Expected_Monthly_Units. Compute for commissioned records. Segment by State and System_Size_kW. Which segment underperforms (<85% ratio)? Plot a heatmap.	Installation quality and generation performance
D4	Calculate the payback period for each commissioned beneficiary under the CAPEX model: Payback_Months = (Market_Cost_Rs - CFA_Sanctioned_Rs) / Monthly_Bill_Savings_Rs. Segment by Beneficiary_Category and plot boxplots. What is the median payback for BPL vs. General households?	Affordability and financial sustainability for beneficiaries

Part 2: SQL Analysis

Import the CSV into any relational database as a table named pmsgmby_applications. Write SQL queries for each of the following. State the SQL dialect used.

Assumed table structure (primary columns):

- pmsgmby_applications (Application_ID, State, DISCOM, Beneficiary_Category, System_Size_kW, Benchmark_Cost_per_kW, Market_Cost_Rs, CFA_Sanctioned_Rs, Application_Date, TFC_Date, Commission_Date, CFA_Credit_Date, TFC_Processing_Days, CFA_Credit_Days_After_Commission, Monthly_Bill_Savings_Rs, Has_Loan, Actual_Interest_Rate_pct, Loan_Approved, Vendor_ID, ALMM_Compliant, Vendor_Rating, Has_Grievance, Grievance_Type, Grievance_Resolved, Grievance_TAT_Days, Satisfaction_Score, DISCOM_Incentive_Received_Rs, DISCOM_Incentive_Utilised_Rs, Special_Category_State, ...)

SQL Questions

Q#	Question	What to Evaluate
S1	Write a query to count applications, commissioned installations, and the commissioning conversion rate (%) by State. Order by conversion rate descending. Include only states with at least 50 applications.	Pipeline conversion; state-level performance ranking
S2	Identify all DISCOMs where the average CFA_Credit_Days_After_Commission exceeds 60 days (double the SLA). Return DISCOM name, state, average days, and count of applications in the breach category.	DBT fund flow delay detection; accountability reporting
S3	Using a window function (RANK or DENSE_RANK), rank vendors within each State by average Vendor_Rating. Return the top 3 vendors per state. Include vendor ID, state, average rating, and number of installations.	Window functions; vendor performance ranking
S4	Write a CTE-based query to compute the month-wise application volume (year-month from Application_Date) and the cumulative total. Identify the month with the highest single-month growth rate.	CTEs; date functions; trend and growth analysis
S5	For grievances: return a summary table with Grievance_Type, total count, resolved count, resolution rate (%), and average Grievance_TAT_Days. Flag any type where resolution rate < 60% or average TAT > 30 days with a label 'Requires Attention', else 'On Track'.	Conditional logic (CASE WHEN); aggregation; SLA compliance flagging
S6	Write a query to compute the DISCOM incentive utilisation gap: DISCOM_Incentive_Received_Rs – DISCOM_Incentive_Utilised_Rs. Return the top 10 DISCOMs by total unutilised	Multi-aggregation; joins within same table using subquery/CTE; business insight

	incentive amount. Join this with the average TFC_Processing_Days for each DISCOM in the same query.	
S7	Segment applications into loan interest rate brackets: $\leq 7\%$ (Compliant), 7–9% (Marginal), $> 9\%$ (Non-Compliant). For each bracket, compute: count, average loan amount, average system size, and state with highest count. Present as a single result set.	CASE WHEN bucketing; multi-level aggregation; financial compliance analysis
S8	Identify 'high-value underserved' beneficiaries: those who are BPL or SC/ST, have a system of 3 kW, did NOT get a loan approved (Loan_Approved = False or null), and whose CFA covered less than 50% of their Market_Cost_Rs. Return count by state and the average out-of-pocket cost.	Multi-condition filtering; equity gap detection; complex WHERE clause

Advanced Excel

Open PMSGMBY_Dataset.csv in Excel. Perform the following tasks. Each task should be on a clearly labelled sheet. Candidates are required to share the excel file names as “Candidate Name_Assignment_2”

Excel Questions

Q#	Question	What to Evaluate
E1	Sheet: 'Pivot_State_Summary' — Create a Pivot Table showing by State: Total applications, % Commissioned, Average Monthly_Bill_Savings_Rs, Average Satisfaction_Score, and ALMM Compliance %. Apply conditional formatting to highlight states where % Commissioned < 60% in red.	Pivot tables; conditional formatting; multi-KPI summary
E2	Sheet: 'CFA_Check' — Use IF, MIN, and formula logic to validate CFA_Sanctioned_Rs against scheme rules. Add a column Calculated_CFA that computes what CFA should be based on System_Size_kW and Benchmark_Cost_per_kW. Add a column CFA_Error = Calculated_CFA – CFA_Sanctioned_Rs. Flag any rows where CFA_Error > 500.	Complex formula logic; data validation; scheme rule compliance check
E3	Sheet: 'Pipeline_Dashboard' — Build an Excel dashboard with 3 charts: (a) Bar chart of average pipeline stage days by top 10 states by volume, (b) Pie chart of Grievance_Type distribution, (c) Line chart of monthly application volume trend. Use slicers linked to Beneficiary_Category.	Dashboard design; chart creation; slicer interactivity
E4	Sheet: 'Vendor_Analysis' — Use SUMIF, COUNTIF, and AVERAGEIF to build a vendor-level summary without pivot tables: for each unique Vendor_ID compute total installations, average Vendor_Rating, ALMM compliance count, and % installations with grievances. Sort by total installations descending.	SUMIF / COUNTIF / AVERAGEIF; formula-based aggregation without pivot
E5	Sheet: 'Loan_Compliance' — Identify all rows where Has_Loan = TRUE and Actual_Interest_Rate_pct > 7. Use VLOOKUP or INDEX-MATCH to pull in the State name (it is already in the data but simulate a cross-sheet lookup from a reference table you create on Sheet 'State_Ref'). Count violations per state using COUNTIFS.	VLOOKUP / INDEX-MATCH; COUNTIFS; multi-sheet referencing
E6	Sheet: 'Payback_Model' — Create a payback period calculator: Payback_Months = (Market_Cost_Rs – CFA_Sanctioned_Rs) / Monthly_Bill_Savings_Rs. Add a column Payback_Category: 'Short' (≤36 months),	Formula-based categorisation; nested IF; payback analysis

	'Medium' (37–72), 'Long' (>72 months). Build a pivot showing count and average payback by Beneficiary_Category and Payback_Category.	
E7	Sheet: 'Macro_Report' (Optional — Bonus) — Record or write a VBA macro that: (a) auto-fits all column widths, (b) applies a standard header format (bold, blue background, white font) to row 1, (c) exports the current sheet as a PDF named with today's date.	VBA basics; automation; professional output formatting